

Causal Localization of the Refusal Direction in Audio Language Models

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Abstract

A large audio language model (LALM) attaches a speech front end to a text language model (LM) that is already safety-aligned. When such a model refuses a harmful spoken request, is the refusal carried by the front end, or inherited from the text LM? We test this with causal interventions. At each model’s audio-to-LM interface and at tested LM residual layers, we fit a direction separating harmful from benign prompts, ablate its component, and measure the resulting change in the model’s first-token refusal margin. Four of the five models are evaluated under held-out category shift. Across five LALMs spanning three backbone families, with three models passing a baseline safety gate, the largest tested effects occur in a mid-to-late LM band, while ablations of the tested interface directions have little effect. On Qwen2.5-Omni, ablating the L16 direction changes the margin by -7.10 , versus -0.013 at the projector. The audio pathway is still in use: zeroing the encoder output changes the margin by -4.7 . In the same model, the contrast is linearly decodable at an early layer where single-layer ablation has little effect, and a direction fitted on the LM backbone alone transfers to the full audio model. Because harmful and benign prompts also differ in form, we interpret the direction as refusal-linked rather than harmfulness-specific. These interventions localize dependence of the refusal margin, not where refusal is computed. Moving the margin also does not always change what the model writes. Safety audits of these models should use interventions rather than rely on probes alone, and should examine the inherited text LM alongside the audio interface.

Index Terms: audio language models, refusal direction, mechanistic interpretability, causal localization, AI safety

1. Introduction

Large audio language models (LALMs) attach a speech front end to a text LM that has already been safety-aligned. That architecture leaves a practical ambiguity: when harmful speech raises refusal logits, is the relevant direction carried by the audio pathway, or by the inherited LM? The question matters because speech inputs are a common weak point for LALM safety [1, 2, 3, 4, 5, 6]. Throughout, we measure refusal with one number, $G(x)$: the logit margin between refusal and compliance tokens at the first output position (§2.4). We operationalize the question by asking where removing a harmful-versus-benign direction makes $G(x)$ fall.

In our experiments, harmful/benign information is decodable from many activations, but decodability alone does not establish causal use [7, 8, 9]. Recent work studies refusal geometry and cross-modal safety dynamics in audio and multi-modal LMs, including alignment and inference-time interventions [10, 11, 12, 13, 14, 15]. To our knowledge, this literature lacks a fixed held-out, site-by-site test separating where the contrast is decodable from where removing it changes $G(x)$.

We therefore use direct causal tests. At the audio interface and at LM residual sites, we fit a rank-1 direction separating harmful from benign inputs on training categories, then remove it under held-out category shift. The refusal margin depends most strongly on a mid-to-late LM band, while the tested interface directions have little effect. Zeroing the encoder output nevertheless shows that the audio pathway is still in use. On Qwen2.5-Omni, a direction fitted on the LM backbone alone transfers to the full audio model. These interventions localize causal dependence of the first-token margin, not where refusal is computed or how generation is controlled.

2. Methodology

Unless otherwise noted, intervention effects are reported as changes in the first-token margin $G(x)$ (§2.4); higher G means more refusal.

2.1. Datasets and models

Data. The probe set is 588 matched harmful/benign *audio* pairs spanning 14 harm categories. Harmful prompts are drawn from three standard text safety benchmarks: AdvBench [16], HarmBench [17], and JailbreakBench [18], deduplicated within category.¹ Each prompt is rendered to speech by a fixed text-to-speech (TTS) voice, so the model receives the harmful content as audio. Every clip is a *verbatim* reading of a text prompt, so transcription-mediated transfer remains a standing caveat; more generally, text priors can obscure whether performance truly relies on audio [19, 20] (§4).

Hard negative. Each harmful prompt is paired with a *synthesized* domain-matched hard negative: a benign prompt about the same harm domain, for example “how incident responders contain malware” opposite a malware-authoring request, generated from a fixed template bank and rendered through the same TTS pipeline. This pairing matches harm domain and reduces coarse topic confounding. Its cost is a residual prompt-form difference, templated benign against natural-imperative harmful, whose footprint we measure directly (§3.1) and discuss (§4).

Held-out. Two held-out protocols guard against direction-fit circularity: a *group-blocked* split that holds out four entire unseen harm categories, and *leave-one-group-out* (LOGO) over 13 of the 14 categories.² Directions are fit on the training side of a split and evaluated on the held-out side.

Models and sites. Qwen2.5-Omni-7B [21] (28 LM decoder layers) is primary. Peers are Audio Flamingo 3 [22] (same Qwen2.5-7B backbone, different Whisper variant and projec-

¹498, 79, and 11 harmful prompts respectively (588 total; 1176 clips with their matched benign hard negatives), after dropping within-category near-duplicates at token-Jaccard ≥ 0.8 .

²Group-blocked holds out drugs, cyber, bio.chem, and fraud.scam ($N=238$ pairs); LOGO aggregates 13 folds ($N=584$), omitting drugs ($N=4$, too small for its own fold).

tor), Voxtral-Mini-3B [23] (30-layer Ministral), and Ultravox-v0.4.1 [24] and DeSTA2.5 [25] (32-layer Llama-3.1); DeSTA holds the LM *frozen* and feeds it a Q-Former audio interface plus a spliced ASR transcript. Together the five LALMs span three LM-backbone families. We probe the relevant front-end interface and LM residual outputs. Qwen2.5-Omni and AF3 use $\ell \in \{8, 14, 16, 20\}$; peers use denser early-to-late grids (8, 9, and 9 LM sites for Voxtral, Ultravox, and DeSTA), compared by fractional LM depth. Interface hooks test selected directions or subspaces at each model’s probed audio-to-LM interface, not the full audio front end, whose overall contribution is assessed separately.

2.2. Refusal direction and decodability

Following Ardit et al. [26], let $\mathbf{h}_i^{(\ell)} \in \mathbb{R}^d$ be the residual-stream activation for prompt i at layer ℓ and the *readout position*: the last position of the chat-templated prompt, whose next-token logits define $G(x)$ (§2.4). With $\mathcal{H}$ and $\mathcal{B}$ the harmful and benign *training* prompts (audio forwards), the per-layer direction is the unit-normalized difference of means

$$\mathbf{r}^{(\ell)} = \frac{\boldsymbol{\mu}_{\mathcal{H}}^{(\ell)} - \boldsymbol{\mu}_{\mathcal{B}}^{(\ell)}}{\|\boldsymbol{\mu}_{\mathcal{H}}^{(\ell)} - \boldsymbol{\mu}_{\mathcal{B}}^{(\ell)}\|_2}, \quad \boldsymbol{\mu}_{\mathcal{S}}^{(\ell)} = \frac{1}{|\mathcal{S}|} \sum_{i \in \mathcal{S}} \mathbf{h}_i^{(\ell)}, \quad (1)$$

fit independently per layer and per training fold. At the front-end projector, which has no comparable text readout position, $\mathbf{r}$ is instead the unit-normalized difference of means over audio-frame positions (mean-pooled per clip). The rank- k projector check uses this axis plus the top $k-1$ principal components of training harmful projector residuals projected onto $\mathbf{r}^\perp$; the resulting basis is orthonormalized and ablated as one subspace. The audio-derived direction is used throughout. For the cross-modal transfer analysis (§3.2) we also fit a text counterpart: the same prompts are passed as text tokens to the LM backbone, bypassing the speech front end, with the direction taken at the same readout position.

Decodability probe. To separate “decodable” from “causally used,” we project evaluation prompts onto $\mathbf{r}^{(\ell)}$ and compute harmful-vs-benign AUROC of that scalar (pair-level 70/30 train/eval split, in-distribution not group-blocked; eval $N=177$ pairs; LM layers only). AUROC is rank-based, so offsets and scales of raw projections do not matter. Decodability is correlational by construction; the causal protocol is §2.3–2.4, and §3.3 shows the two dissociate.

Because harmfulness and refusal need not share a direction [27, 28], and our benign prompts differ in form (§2.1), we read $\mathbf{r}^{(\ell)}$ as a refusal-linked harmful-vs-benign contrast rather than a harmfulness-pure axis (§4).

2.3. Causal intervention

Three operations act on a hidden state $\mathbf{x}$ at the hooked module output:

$$\text{noop}(\mathbf{x}) = \mathbf{x}, \quad \text{ablate}(\mathbf{x}; \mathbf{r}) = \rho \mathbf{x}_\perp, \quad \text{steer}(\mathbf{x}; \mathbf{r}, k) = \mathbf{x} + k\mathbf{r}, \quad (2)$$

where $\mathbf{x}_\perp = \mathbf{x} - (\mathbf{x}^\top \mathbf{r})\mathbf{r}$ removes the rank-1 component along $\mathbf{r}$ and $\rho = \|\mathbf{x}\|/\|\mathbf{x}_\perp\|$ rescales to preserve the activation norm. NOOP is a plumbing control, ABLATE the necessity test, and STEER the constructive companion. Ablation is applied at *every* token position, while $\mathbf{r}$ is fit at the readout position only: a last-token direction applied throughout the sequence. The reported add-to-benign test uses one training mean-shift scale

($\gamma=1$), i.e. $k = \|\boldsymbol{\mu}_{\mathcal{H}} - \boldsymbol{\mu}_{\mathcal{B}}\|_2$. A fourth operation, *scalar rescue*, records the prompt’s readout-position pre-ablation coordinate $a_i = \mathbf{x}_{i,\text{noop}}^\top \mathbf{r}$, ablates, then adds $a_i \mathbf{r}$ back at all positions; recovery is $(G_{\text{patch}} - G_{\text{ablate}})/(G_{\text{noop}} - G_{\text{ablate}})$ per prompt.

To *localize*, we ablate each candidate layer *independently* (one hooked layer per run), so a site’s effect estimates how the held-out margin depends on removing that layer’s fitted direction. This differs from Ardit et al. [26], who select a direction at ℓ^* and ablate it at every layer. Independent ablation asks where the held-out margin depends on the direction; §3.3 additionally tests joint ablation of L08–L15.

2.4. Readout and localization protocol

A chat model that refuses almost always begins its answer with an opener such as “I’m sorry” or “I cannot”, and one that complies begins with an opener such as “Sure” or “Here”. We therefore read the model’s lean toward refusal off its first output token. Let $\mathcal{R}$ be the token IDs of the refusal openers and $\mathcal{C}$ those of the compliance openers;³ $G(x)$ is the log-sum-exp margin between the two sets on the first-token logits $z_t(x)$:

$$G(x) = \log \sum_{t \in \mathcal{R}} e^{z_t(x)} - \log \sum_{t \in \mathcal{C}} e^{z_t(x)}, \quad (3)$$

so that higher G means a stronger lean toward refusal. The two word lists were fixed before any intervention was run and are the same for every model; only the token IDs are re-derived per tokenizer. $\mathbf{l}$ is on the refusal list (and $\mathbf{l}$ ’s shares its first token) because that is how these models begin a refusal: in the cached Qwen2.5-Omni generations, all 300 refusals open with “I’m”, and none of the 180 compliant answers—including the 60 that ablation induced—begins with $\mathbf{l}$. Section 4 reports how well the sign of G agrees with a full-text judge. In the results, $\Delta G = G_{\text{ablate}} - G_{\text{noop}}$ is the change on held-out harmful clips, paired per clip, and N counts matched pairs in the split; benign, pairwise, and behavioral results are labeled as such.

Specificity nulls. To check the effect is specific to $\mathbf{r}$ and not a generic consequence of perturbing the residual stream, we ablate K random unit directions identically at every site⁴ and compare their mean effect with that of $\mathbf{r}$. A stricter pre-specified structure-matched null (a direction fit on benign-vs-benign data, same geometry but no harm contrast) is reported in §4.

Localization and claim level. For Qwen2.5-Omni we use $|\Delta G| \geq 3.55$, half the magnitude of the initial group-blocked L16 effect (7.10), as a descriptive threshold for the “causal band”; dense follow-ups map its extent (§3.1). The primary readout is always first-token $G(x)$, a logit-level proxy. Behavioral results use greedy 512-token generations scored by the HarmBench classifier `cais/HarmBench-Llama-2-13b-cls` [17], are labeled separately, and do not follow from logit control (§3.4, §4). Every unlabeled ΔG number is logit-level.

³Full frozen sets (each token with and without a leading space): refusal = {I, I’m, Sorry, Unfortunately, As, Cannot, No, Apolog}; compliance = {Sure, Here, Certainly, Okay, To, Yes, Of}.

⁴ $K=4$ for the held-out sweeps, 16 for the peer sweeps, 32 for the AF3 check, and 2 for the rescue, add-induce, and cross-modal analyses.

Table 1: *Qwen2.5-Omni held-out ablation effects.* ΔG on harmful clips with paired-bootstrap 95% CIs, under group-blocked ($N=238$) and leave-one-group-out ($N=584$) splits; random-direction nulls satisfy $|\Delta G| \leq 0.02$ at every site. †: dense group-blocked follow-up.

Intervention	Group-blocked	LOGO
Projector, rank-1	−0.013 [−0.033, 0.005]	−0.005 [−0.018, 0.008]
Projector, rank-16 subspace	−0.14 [−0.21, −0.08]	—
LM L08	−0.10 [−0.12, −0.09]	−0.12 [−0.14, −0.11]
LM L14	−2.51 [−2.64, −2.39]	−3.29 [−3.40, −3.19]
LM L16	−7.10 [−7.28, −6.93]	−7.53 [−7.67, −7.38]
LM L17†	−9.19 [−9.38, −8.99]	—
LM L20	−5.43 [−5.56, −5.27]	−4.96 [−5.08, −4.82]
LM L16, natural-benign refit	−1.79 [−1.88, −1.70]	—

3. Results

3.1. The margin depends on a mid-LM direction, not on the tested projector direction

Removing a single direction at a mid-LM layer sharply lowers the refusal margin; removing the analogous direction at the projector output does not (Table 1, Fig. 1). At layer 16 of Qwen2.5-Omni, ablation changes the held-out margin by $\Delta G = -7.10$, against -0.013 at the projector, comparable to the random-direction null. LOGO replicates this contrast; L08 remains small; and the rank- k projector test of §2.2 reaches at most 2% of the L16 effect at $k = 16$. Within the projector subspaces we tested, no removal affects the margin comparably.

The effect is not confined to L16 but spread over a band of layers: it rises from L14, peaks at L15–L18 with its maximum at L17, and is still about half its maximum at L26 of the model’s 28 layers (Table 1, Fig. 1). We use L16 below as a representative of this band. Across the other four models the largest tested effects also occur at LM sites, with model-dependent mid-to-late profiles, including a broad Voxtral plateau and a DeSTA peak at L20 (§3.4).

The audio channel is open: zeroing the encoder output changes G by -4.7 ($N=16$ clips). The projector null is narrower, covering only the directions and subspaces we tested at that site. Removing a direction inside the encoder also leaves G unchanged, but only 2–3% of the removed component survives the encoder’s final normalization and pooling, so that null is uninformative (§4).

The size of the L16 effect depends on the benign set but stays far larger than at L08. The headline -7.10 compares harmful prompts phrased as natural imperatives with benign prompts generated from templates, so part of it may reflect prompt form rather than harmful content. Refitting the direction with natural benign prompts from XSTest [29] reduces the L16 effect to -1.79 , below this causal-band threshold of $|\Delta G| \geq 3.55$ (§2.4). We therefore treat -7.10 as an effect of the matched design rather than of harmful content alone. The ordering survives the refit: $|\Delta G|$ remains far larger at L16 than at L08 ($\Delta G = -0.03$), and random directions again leave G unchanged.

3.2. Readout-level rescue and text/audio transfer

One scalar coordinate along the L16 direction recovers most of what ablation removes. Scalar rescue (§2.3) records each prompt’s coordinate along $\mathbf{r}^{(16)}$ before ablation and adds it back afterwards. On the group-blocked split of §3.1, ablation carries the mean margin below zero (6.85 to -0.25), and the mean per-prompt recovery is 0.950, against 0.000 for a random coordinate (Table 2). In the opposite direction, adding $\mathbf{r}^{(16)}$ to benign

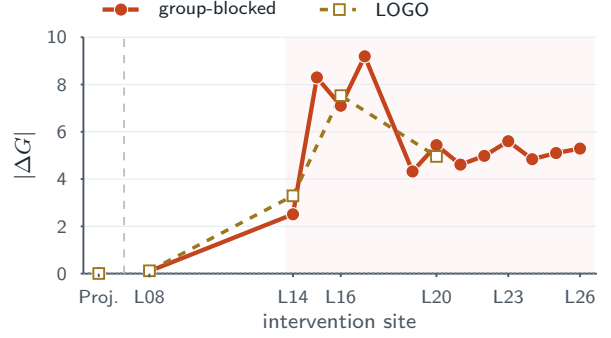


Figure 1: *Ablation localizes the margin effect to an LM band.* Held-out $|\Delta G|$ at each intervention site, under group-blocked ($N=238$) and LOGO ($N=584$) category shift; shading marks L14–L26.

Table 2: *Readout-level rescue and transfer at L16 (Qwen2.5-Omni).* Group-blocked split unless noted. Transfer rows ablate a direction fitted in one modality during the other modality’s forward pass, and give the result as a fraction of that pass’s own effect; the paraphrase row is a fraction of the verbatim effect on the same 47 items.

Test at L16	Value	Random control
Scalar rescue (recovery)	0.950 [0.947, 0.952]	0.000 [−0.001, 0.001]
Add $\mathbf{r}^{(16)}$ to benign (ΔG)	+7.24	−0.24
Text-fit → audio pass	0.99	—
Audio-fit → text pass	0.73	—
Paraphrased audio ($N=47$)	0.75 [0.65, 0.84]	—

activations raises the benign margin by 7.24.

A direction fitted on the LM backbone alone transfers to the full audio model. The text-fitted direction, ablated in the audio pass, reproduces 0.99 of the effect of the audio-fitted direction at L16; in the reverse direction the audio-fitted direction reproduces 0.73 of the text-fitted effect; the two have cosine similarity 0.76 (Table 2). The transfer is substantial but incomplete: it is asymmetric, and the cosine is well below one. To check whether shared wording carries the transfer, we paraphrased the prompts in a separate language-model context, keeping only those a separate judging pass rated at least 4/5 for intent preservation and whose token-level Jaccard overlap with the original was below 0.3; the direction was fit on 541 disjoint pairs and evaluated on 47 paraphrased clips. The effect is still 0.75 of the verbatim-audio effect and remains specific to the fitted direction (mean random-direction $|\Delta G| \approx 0.045$). Shared wording alone therefore cannot fully explain the transfer. Because all main clips are verbatim TTS readings, however, the model may still reuse the text feature through an internal transcript of the speech (§4).

3.3. Decodability appears earlier than single-layer causal dependence

A contrast can be easy to decode at a layer where removing it does almost nothing. The harmful-vs-benign contrast is nearly perfectly decodable at every LM site we intervene on (AUROC ≥ 0.997), yet ablating it at L08 changes G by only -0.103 , roughly 1/70 of the L16 effect. Decodability alone therefore does not localize causal dependence; the templated form of the benign prompts likely contributes to the ceiling (§4).

Jointly, however, layers L08–L15 do change the margin. Removing the L08–L15 directions together, each layer with

Table 3: **Cross-model summary.** *H-refuse is the baseline harmful-refusal rate; the gate requires it ≥ 0.70 with answer-benign ≥ 0.80 , and $\dagger$ marks models that fail it. Harm flip is the fraction of baseline refusals ablation turns into harmful answers. ΔG is held-out group-blocked ablation, except AF3, whose in-sample fit makes its magnitude incomparable; Baseline G and Harm flip come from a separate behavioral run.*

Model	H-refuse	Baseline G	LM site (ΔG)	Harm flip
Qwen2.5-Omni	1.00	6.85	L16 (−7.10)	0.75
AF3 $\dagger$	0.68	2.97	L20 (−4.22)	0.24
Voxtral $\dagger$	0.60	4.08	L18 (−5.56)	0.49
Ultravox	0.83	13.12	L20 (−7.69)	0.10
DeSTA	0.87	8.18	L20 (−6.03)	0.00

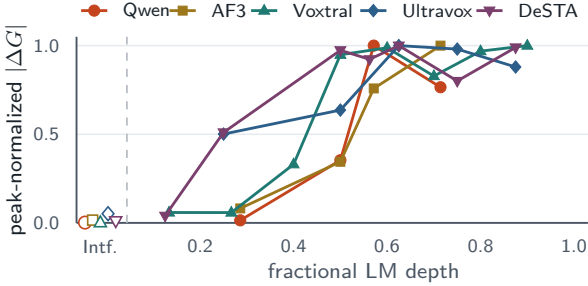


Figure 2: **Layer-wise localization across models.** For each model, $|\Delta G|$ is normalized by its peak over the tested LM sites and plotted against fractional LM depth; absolute magnitudes are therefore not comparable across models. Open markers left of the divider are the tested audio interfaces; their horizontal positions carry no depth information.

its own training-fit direction on the same group-blocked split, changes the held-out margin by -6.01 (CI $[-6.24, -5.78]$), or 0.85 of the single-site L16 effect in the same run (-7.07 ; joint random null -0.01 ± 0.07). One reading is self-repair [30, 31, 32]: when a single layer’s direction is removed, later layers may restore the margin. But this interval includes L15, whose own effect is large (Fig. 1), so this control does not isolate that reading. Either way, single-layer ablation localizes margin dependence, not refusal computation.

3.4. Cross-backbone replication and the limits of logit control

The same LM-band pattern appears in models from all three backbone families (Table 3, Fig. 2). Three of the five localized models pass our pre-specified safety gate; AF3 [22] and Voxtral do not (Table 3). A sixth model, LLaMA-Omni [33], refused almost nothing at baseline and was excluded before localization. The AF3 direction is also fit in sample on the 41 prompts it refuses, so its magnitude is not comparable with the held-out values.

The contrast of §3.1 between LM sites and the tested interface recurs in the other models: ablation moves the margin by -0.004 at the Voxtral connector, by $+0.40$ at the Ultravox projector against -7.69 at L20, and by -0.07 at DeSTA’s Q-Former perception output. The shape of the band differs by family, but in every model the LM sites produce far larger effects than the interface we tested.

Moving the margin does not always change what the model writes. Across the four held-out models, band-site ablation turns 0.75 of Qwen’s baseline refusals into harmful an-

swers (0.00 for a random direction), compared with 0.49 for Voxtral, 0.10 for Ultravox, and 0.00 for DeSTA, even though the ablation moves the margin by a similar amount in all four (Table 3). Higher baseline margins in Ultravox and DeSTA may contribute to their smaller harm-flip fractions. However, Table 3 reports behavioral baselines and localization-run ΔG from separate runs, so it does not establish post-ablation margins for the behavioral examples.

4. Limitations and Discussion

We do not claim that refusal is computed in the LM. We show where the first-token margin $G(x)$ falls when a single direction is removed; where refusal is computed, and whether moving the margin controls what the model generates, are separate questions.

$G(x)$ reads only the first token, and a first token that leans toward compliance does not guarantee a harmful answer: the fraction of refusals that ablation turns into harmful answers ranges from 0.75 to 0.00 across models (Table 3), all scored by one HarmBench classifier. On Qwen2.5-Omni, Claude Opus 4.8 labeled the 120 unablated outputs and the HarmBench classifier the 60 ablated harmful outputs. The sign of G agreed with these labels on 0.70 of the ablated outputs, whose margins lie near zero; on 0.81 of all 180 items (Cohen’s $\kappa = 0.62$); and on 0.97 of items with $|G| > 3$. This mixed-evaluator check has not been repeated on the other models.

The projector null applies to the tested directions and subspaces, not to the audio pathway. Zeroing the whole audio encoder moves G by -4.7 . No single direction fit on mean-pooled projector frames, and no rank- k subspace up to $k=16$, changes G by more than 2% of the L16 effect when removed. Distributed, position-specific, or nonlinear mechanisms in the audio pathway are not ruled out [34].

A pre-specified structure-matched specificity check was ambiguous under its decision rule: across eight benign-versus-benign splits, mean $|\Delta G|$ was 0.47 (maximum 1.89), above the isotropic random-direction band but small relative to the fitted-direction effect. For the VITS and SpeechT5 voices, XSTest-safe prompts yield $\Delta G = -2.86$ and -2.93 , versus -6.75 and -7.58 on the corresponding unsafe contrasts. Together these controls argue against a harmfulness-pure reading: we interpret the direction as refusal-linked rather than harmfulness-specific.

All speech is synthetic: on Qwen2.5-Omni the contrast between L16 and the projector reproduces across three further TTS voices, but human-recorded speech remains untested. The main next steps are to test human-recorded speech and audio mechanisms not mediated by a transcript.

5. Conclusion

We asked whether refusal in audio language models is carried by the speech front end or by the inherited language-model backbone. Under the tested rank-1 intervention, the first-token refusal margin $G(x)$ depends most strongly on a mid-to-late LM residual band, while the tested projector/connector directions stay near zero. This localizes dependence of the refusal read-out, not where refusal is computed; it does not make the audio pathway irrelevant. For safety auditing, decodability alone does not establish causal use; inherited text-LM backbones should be audited alongside audio interfaces.

Ethics statement. We report aggregate white-box safety measurements only and release no fitted directions, prompts, or generations.

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